MATLAB Projects based on Digital Communications Applications

Oct 15th 2023

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Albert Einstein in his own words said that,

“Education is what remains after one has forgotten what one has learned in school.”

Electronic and Communications Engineering

University of Kufa

Broadband scope of course project

Students will learn how to model applications in communication systems using MATLAB

- Underhand a mathematical model for an system or sub-system & use MATLAB to perform
  - Calculation
  - Plotting
  - Analysis possibly
  - Sanity check

The General representation of signals

Does the signal propagation in transmission media digitally (0s & 1s)? For example in optical cables how signals propagate?

The trend of using digital communications is to overcome the limitations of using analog communications

Analog Communication issues

- Limited BW
- High Losses
- High Distortion
- Less reliable
- Complicated design & high cost
- Less implementation flexibility
- No encryption or advanced coding
- Do Not support spread spectrum technique

In short words, the digitized signals are used to obtain an efficient exchange of information with less losses and distortion than analog communication and higher BW over a long distance

Historical Background: Three key points that contributes to develop communication system

- Information theory & Coding
- Internet (networking)
- Wireless communications

Impact

Digital communications in revolutionary ways

Ref: [2]

Elements of a communication system

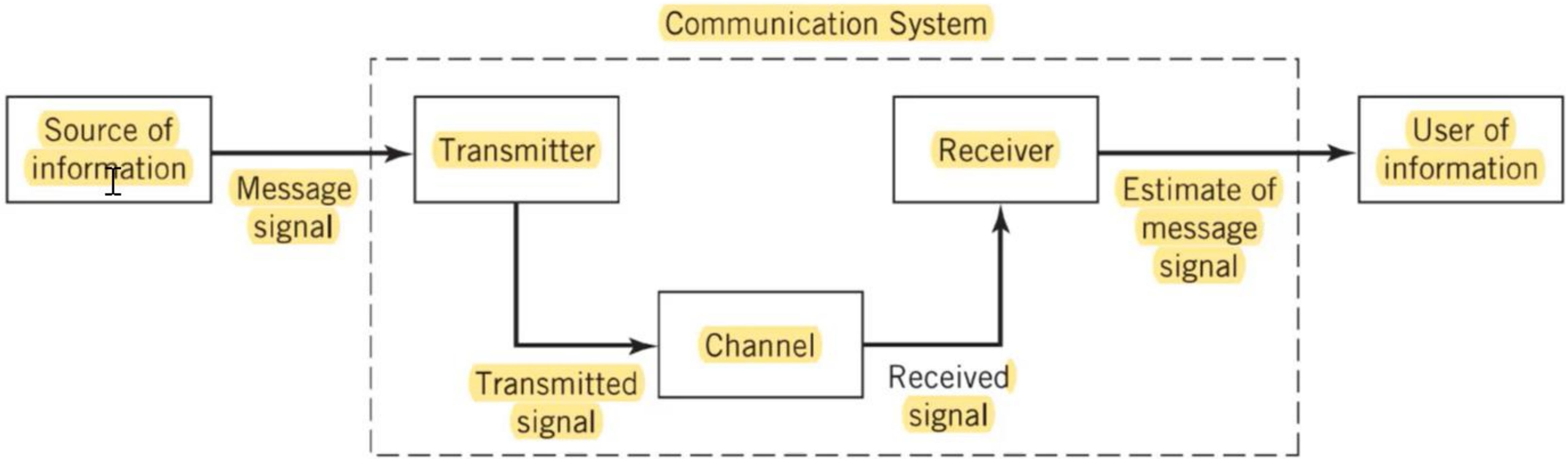


Figure 1.1 Elements of a communication system.

Course Project Requirements---10 points

- Oral Presentation ---present template will be provided
  - Progress presentation
  - Final presentation

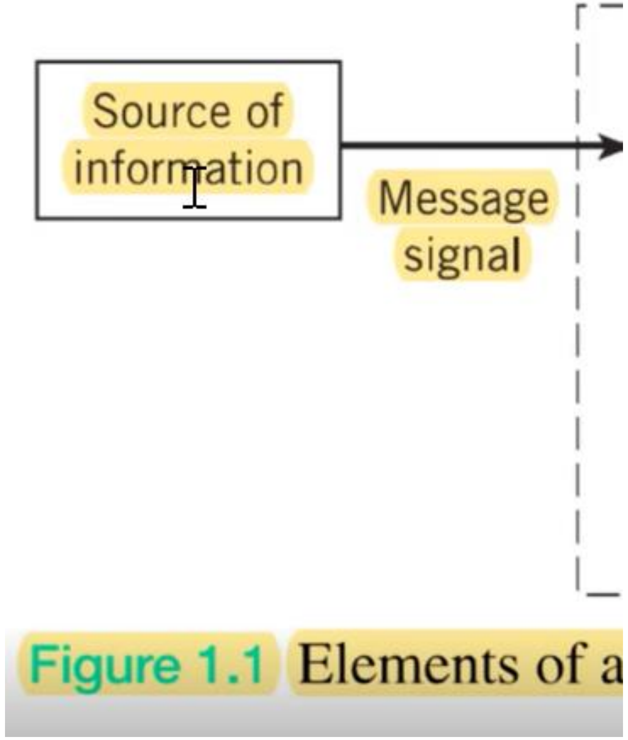
- Report ---Report template will be provided

- Published code of your MATLAB work

<u>Other requirements</u>

- All students must present to avoid failure in the project
- Use references and avoid plagiarism
- Submit their progress prevention on the assigned due date
- Build their MATLAB code using cells. units, and in-line descriptions

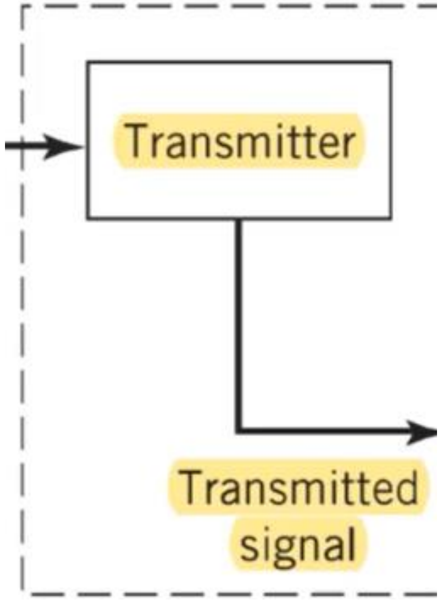
Project Topics



Signal Generation and Visualization

periodic and aperiodic waveforms, swept-frequency sinusoids, and pulse trains

Project Topics



A. Pulse-amplitude modulation

B. PSK (phase-shift keying)

C. FSK (frequency-shift keying)

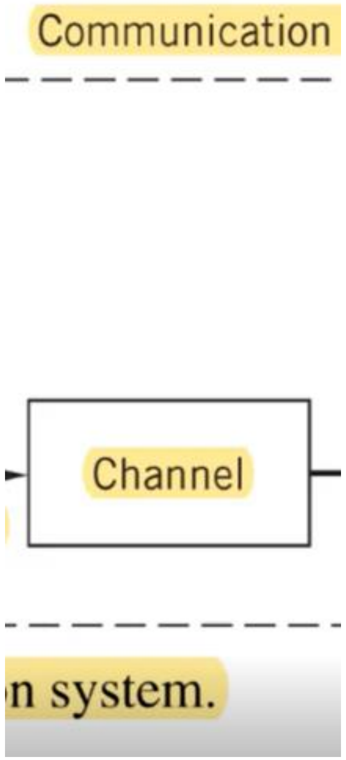
D. ASK (amplitude-shift keying)

E. QAM (quadrature amplitude modulation)

Elements of a communication system

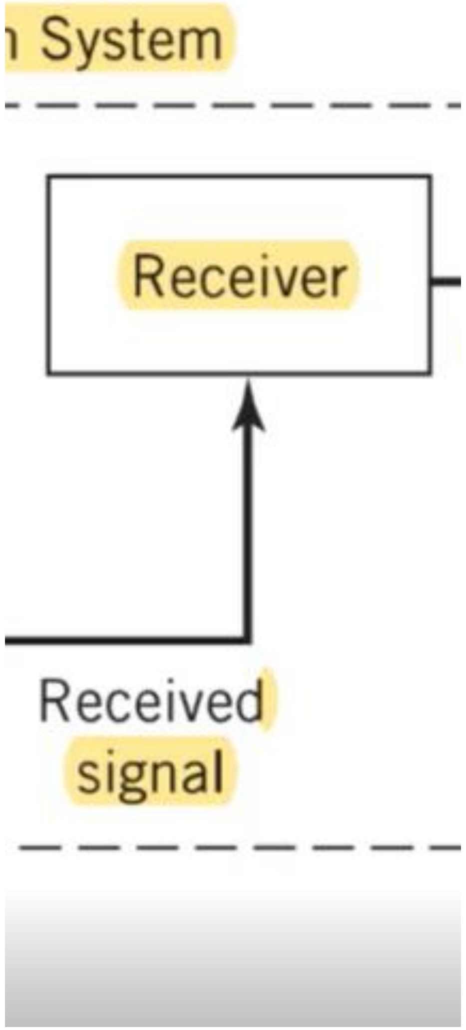
11. Digital filtering

12. Add white Gaussian noise to signal with scatterplot



Elements of a communication system

- Received signal visualization in scope and spectrum analyzer
- Generate eye diagram using MATLAB



Project Topics

Speaker Identification Using Pitch and MFCC

Mel Frequency Cepstrum Coefficient (MFCC): MFCC is designed to model features of audio signal

Refs:

[1] https://www.mathworks.com/help/audio/ug/speaker-identification-using-pitch-and-mfcc.html

[2] https://www.sciencedirect.com/topics/computer-science/cepstral-coefficient

[3] Mel Frequency Cepstral Coefficient and Its Applications: A Reviewhttps://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9955539

Elements of a communication system

Design a noise cancellation algorithm to remove unwanted noise from audio signals, employing techniques like adaptive filtering

Refs:

[1] https://www.mathworks.com/help/dsp/adaptive-filters.html

[2] https://www.academia.edu/27203825/Noise_Cancellation_Using_an_Adaptive_Filtering_Technique

Elements of a communication system

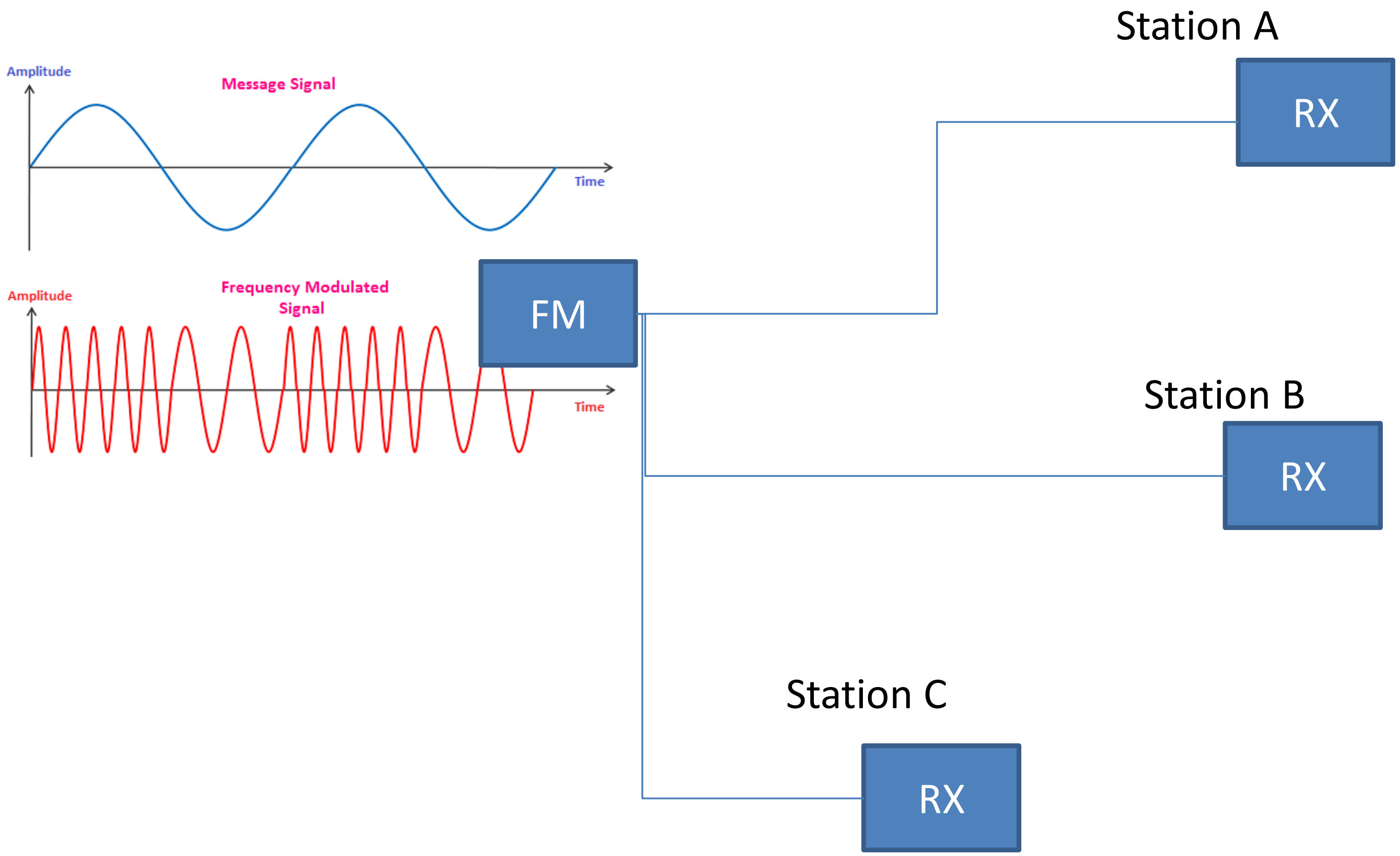
Wireless Communication Channel Modeling:

Develop models to simulate wireless communication channels (e.g., Rayleigh fading, multipath propagation) for performance evaluation and optimization of wireless communication systems

Understand the characteristics of different wireless propagation environments (e.g., urban, rural, indoor) and their impact on channel behavior

Implement algorithms to estimate key channel parameters such as path loss

HW: Write a MATLAB code to generate an FM signal



- Introduce Noise and different delay per path
- Plot your TX and RX signals at different stations
- Remove noise form signals and plot your results in time and frequency domain using fft
- Preform alignment, build your function using find peaks and shifting
- Compare signals after processing

References

[1] https://www.tutorialsmate.com/2021/11/difference-between-data-and-information.html

[2] https://physics.stackexchange.com/questions/314695/electromagnetic-wave-vs-electric-current#:~:text=Electric%20current%20is%20the%20movement,can%20propagate%20through%20a%20vacuum.

A network has a number of stations that are connected with each other through nodes

- Signal Generation and Visualization
- https://www.mathworks.com/help/signal/waveform-generation.html
- Add white Gaussian noise to signal

https://www.mathworks.com/help/comm/ref/awgn.html

- Signal Visualization and Measurements in MATLAB
- https://www.mathworks.com/help/dsp/ug/signal-visualization-and-measurements-in-matlab.html
- Design and simulate streaming signal processing systems
- https://www.mathworks.com/products/dsp-system.html#:~:text=DSP%20System%20Toolbox%20provides%20algorithms,%2C%20IoT%2C%20and%20other%20applications.
- Digital filtering
- https://www.mathworks.com/help/matlab/ref/filter.html